

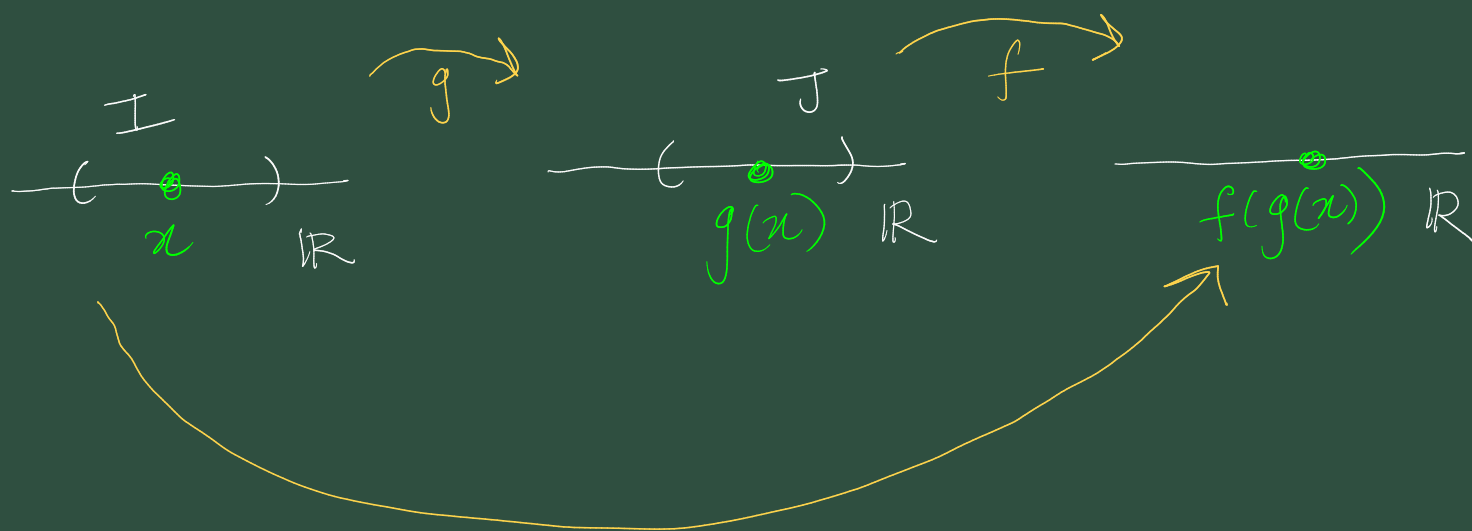
Chain rule  
quotient rule.

chain rule

Consider.  $y = f \circ g$

ie.  $y(x) = f(g(x))$ .

How does  $y'(x)$  depend on  $f, g$   
and their derivatives  $f'(x), g'(x)$ .



$$y = f \circ g$$

Chain rule says, assuming  $g$  is differentiable at  $x$  and  $f$  is differentiable at the point  $g(x)$  (and both are defined in intervals  $I, J$  containing these points, respectively) then the derivative of the composition  $y = f \circ g$  is given by

$$y'(x) = f'(g(x)) g'(x)$$

example

Consider  $y(x) = \ln(4x^3 - 2x^2)$ .

defined for suitable  $x$ .

We see  $y$  is a composition

$$\begin{aligned} y &= \ln(g(x)) \\ &= (\ln \circ g)(x) \end{aligned}$$

where  $g$  is the poly. function

$$g(x) = 4x^3 - 2x^2$$

Applying the chain rule, we get

$$\begin{aligned} y'(x) &= \ln'(g(x)) g'(x) \\ &= \frac{1}{g(x)} \cdot g'(x), \text{ using standard} \\ &\quad \text{derivative of } \log. \end{aligned}$$

$$= \frac{1}{4x^3 - 2x^2} (12x^2 - 4x)$$

$$= \frac{12x^2 - 4x}{4x^3 - 2x^2}$$

Coming back soon for proof of chain rule.

But first we'll prove the quotient rule.

Theorem 4.98

$$\text{If } y = \frac{u}{v}, \text{ i.e. } y(x) = \frac{u(x)}{v(x)}.$$

$$\text{then } y'(x) = \frac{v(x)u'(x) - u(x)v'(x)}{v^2(x)}$$

We can prove quotient rule in a nice way using the chain and product rules, after we see a quotient re-expressed using a combination of prod. and composition.

$$\begin{aligned} y = \frac{f}{g} &= f \cdot \left( \frac{1}{g} \right) \\ &= f \cdot (r \circ g), \end{aligned} \quad \begin{array}{c} \downarrow \\ \leftarrow \end{array}$$

where  $r$  is the reciprocal function.

$$r(x) = \frac{1}{x}.$$

To confirm this.

$$\begin{aligned}
& (f \uparrow (r \circ g))(x) \\
&= f(x) \cdot (r \circ g)(x) \\
&= f(x) \cdot r(g(x)) \\
&= f(x) \cdot \frac{1}{g(x)} \\
&= \frac{f(x)}{g(x)} = \left( \frac{f}{g} \right)(x).
\end{aligned}$$


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Lemma 4.4.9 For the reciprocal.  
 $r(x) = \frac{1}{x}$ , its derivative is

$$r'(x) = -\frac{1}{x^2}.$$

Proof  $r'(a) = \lim_{x \rightarrow a} \left( \frac{\frac{1}{x} - \frac{1}{a}}{x - a} \right)$

$$= \lim_{x \rightarrow a} \left( \frac{\frac{a-x}{xa}}{x-a} \right), \quad \text{adding fractions in the num.}$$

$$= \lim_{n \rightarrow a} \left( \frac{a-n}{na(n-a)} \right)$$

$$= \lim_{n \rightarrow a} \left( \frac{-1}{na} \right), \quad \text{cancelling } (n-a)$$

$$= -\frac{1}{a^2}$$

and this is valid for all  $a \in \mathbb{R} \setminus \{0\}$ .

Then the derivative function of  $r$

$$\text{is } r'(x) = -\frac{1}{x^2}$$

Proof of quotient rule.

$$\text{Consider } y = \frac{f}{g} = f \cdot (r \circ g)$$

$$y'(x) = [f \cdot (r \circ g)]'(x)$$

$$= f'(x) (r \circ g)(x) + f(x) (r \circ g)'(x)$$

, by the product rule.

$$= f'(x) \frac{1}{g(x)} + f(x) r'(g(x)) g'(x)$$

, using def. of  $r$ , the chain rule.

$$= f'(x) \frac{1}{g(x)} + f(x) \left( \frac{-1}{g(x)^2} \right) g'(x).$$

, using def. of  $r'$ , i.e.  
lemma 4.4.9.

$$= \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}, \text{ as required}$$

□

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Towards a proof of chain rule.

Linear approximations with error

Consider again the derivative def.

$$f'(a) = \lim_{h \rightarrow 0} \left( \frac{f(a+h) - f(a)}{h} \right)$$

This suggests the existence of an  
"error function"

$$\varepsilon(h) = \frac{f(a+h) - f(a)}{h} - f'(a)$$

and we know  $\varepsilon(h) \rightarrow 0$  as  $h \rightarrow 0$ .  
Now rearrange this to get

$$\underbrace{f(a+h)}_{\text{new value of } f} = f(a) + \underbrace{f'(a)h}_{\text{linear approximation}} + \underbrace{h\varepsilon(h)}_{\text{Err}(h) \text{ associated error}}$$

Very clear graphical interpretation of this in figure 4.4.4.

Note that in the above not only does  $\text{Err}(h) \rightarrow 0$  as  $h \rightarrow 0$

but also  $\frac{\text{Err}(h)}{h} = \varepsilon(h) \rightarrow 0$  as  $h \rightarrow 0$ .

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Alternative definition of the derivative.

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The derivative  $f'(a)$  is the unique gradient that gives a lin. approx. with error  $\text{Err}(h)$

$f(a+h) = f(a) + f'(a)h + \text{Err}(h)$   
such that  $\frac{\text{Err}(h)}{h} \rightarrow 0$  as  $h \rightarrow 0$ .

Chain rule proof Given  $y = f \circ g$ . Find  $y'(a)$  which we'll do by obtaining the suitable "lin. approx with error" expression for  $y(a+h)$  and discovering the chain rule expression  $f'(g(a))g'(a)$  sitting in the gradient position.

We assume the "lin approx with errors" for  $f, g$

$$1) \quad g(a+h) = g(a) + g'(a)h + \underline{h\varepsilon(h)}$$

$$2) \quad f(g(a)+l) = f(g(a)) + f'(g(a))l + l\mu(l)$$

where  $\varepsilon, \mu$  have convergence properties  
 $\underline{\varepsilon(h) \rightarrow 0 \text{ as } h \rightarrow 0}$  and  $\underline{\mu(l) \rightarrow 0 \text{ as } l \rightarrow 0}$ .

Combine 1) and 2) to find  $y(a+h)$ .

$$\begin{aligned} \boxed{y(a+h)} &= f(g(a+h)), \text{ def of } y. \\ &= f(\underbrace{g(a)}_{\text{yellow}} + \underbrace{g'(a)h + h\varepsilon(h)}_l), \text{ using 1).} \\ &= f(g(a)) + f'(g(a)) (g'(a)h + h\varepsilon(h)) \\ &\quad + l\mu(l) \end{aligned}$$

, where  $l = g'(a)h + h\varepsilon(h)$   
by 2).

$$= f(g(a)) + f'(g(a)) g'(a) h + f'(g(a)) h \varepsilon(h) + l \mu(l)$$

$$= y(a) + f'(g(a)) g'(a) h + \text{Err}(h)$$

where  $\text{Err}(h) = f'(g(a)) h \varepsilon(h) + l \mu(l)$

Notice the  $f'(g(a)) g'(a)$  sitting in the expected gradient position  $y'(a)$ . So we just need to confirm that  $\text{Err}(h)$  has the required convergence property namely  $\frac{\text{Err}(h)}{h} \rightarrow 0$  as  $h \rightarrow 0$ .

$$\frac{\text{Err}(h)}{h} = \frac{1}{h} \left( f'(g(a)) \cancel{h} \varepsilon(h) + (g'(a) \cancel{h} + \cancel{h} \varepsilon(h)) \times \mu(g'(a)h + h\varepsilon(h)) \right)$$

$$= f'(g(a)) \varepsilon(h) + (g'(a) + \varepsilon(h)) \mu(g'(a)h + h\varepsilon(h))$$

But note that  $\varepsilon(h) \rightarrow 0$  as  $h \rightarrow 0$   
so  $gl = g(a)h + h\varepsilon(h) \rightarrow 0$   
as  $h \rightarrow 0$

and so  $\mu(h) \rightarrow 0$  as  $h \rightarrow 0$ .

Therefore using algebra of limits theorem  
for function values

$$\frac{\varepsilon(h)}{h} \rightarrow 0 \text{ as } h \rightarrow 0.$$

This all establishes the required  
linear approx with error expression

$$y(a+h) = y(a) + y'(a)h + \varepsilon(h)$$

where  $y'(a) = f'(a) = g'(a)$ .

This is the chain rule.

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Next Wed

Application of The derivative  
to optimisation problems.